

Global solutions for the interpolated stochastic heat and wave equation with a super-linear diffusion term

Le Chen
Auburn University
le.chen@auburn.edu

Nicholas Eisenberg
Auburn University
nze0019@auburn.edu

Friday 3rd June, 2022

Abstract

In this paper we prove the existence of a global solution to the interpolated stochastic heat and wave equation driven by a space-time white noise and with a super-linear non-linear diffusion term.

MSC 2010 subject classification:

Keywords:

Contents

1 Introduction and main result	1
2 Preliminary results for a globally Lipschitz σ	3
2.1 Some moment bounds	4
3 Proof of Theorem 1.1	10
4 Some comments and current progress on a spatially colored noise	11
4.1 Some moment bounds for a globally Lipschitz σ	14
References	21

1 Introduction and main result

In this chapter, we study the interpolated stochastic heat and wave equation (ISHWE) on the whole space $\mathbb{R}^d$,

$$\begin{cases} \left(\partial_t^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2} \right) u(t, x) = I_t^\gamma \left[\sigma(u(t, x)) \dot{W}(t, x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = u_0(x) & \beta \in (0, 1] \\ u(0, \cdot) = u_0(x), \quad \partial_t u(0, \cdot) = v_0(x) & \beta \in (1, 2), \end{cases} \quad (1.1)$$

E:SupLinSPD

where the fractional differential operators ∂_t^β , $(-\Delta)^{\alpha/2}$ and I_t^γ respectively denote the *Caputo derivative*, the *fractional Laplacian* and the *Riemann-Liouville fractional integral operator*. The noise, $\dot{W}$ is taken to be space-time white and the initial data, u_0 and v_0 , are assumed to be

Borel measurable functions which are Bounded on $\mathbb{R}^d$. The nonlinearity, σ , is *locally Lipschitz* in the sense that the following asymptotic relation holds as $|x|, |z| \rightarrow \infty$:

$$|\sigma(x) - \sigma(z)| \leq \sigma_2 |x - z| [\ln_+(|x - z|)]^\delta, \quad (1.2)$$

where $\sigma_2, \delta > 0$ and $\ln_+(z) = \ln(z \vee e)$ for $z > 0$. Further, this implies that as $|x| \rightarrow \infty$,

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta, \quad (1.3)$$

where $\sigma_1 = \sigma(0)$.

For any $T > 0$ with $t \in [0, T]$ and $x \in \mathbb{R}^d$, the solution to (1.1) is understood in the *mild* sense as the solution to the following stochastic integral equation:

$$u(t, x) = J_0(t, x) + I(t, x),$$

where

$$\begin{aligned} J_0(t, x) &= \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) + [Z^*(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases} \\ I(t, x) &= \int_0^t \int_{\mathbb{R}^d} Y(t - s, x - y) \sigma(u(s, y)) W(ds, dy). \end{aligned}$$

The stochastic integral above is a *Walsh integral* (e.g. see [Wal86]) and the fundamental solution is a triple, $\{Z, Z^*, Y\}$, and each member of the triple is defined through the Fox-H function, however, one can more compactly define them through their Fourier transforms [CHN19, Theorem 4.1]:

$$\begin{aligned} \mathcal{F}Z(t, \cdot)(\xi) &= t^{\lceil \beta \rceil - 1} E_{\beta, \lceil \beta \rceil} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right), \\ \mathcal{F}Z^*(t, \cdot)(\xi) &= E_{\beta, 1} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right), \\ \mathcal{F}Y(t, \cdot)(\xi) &= t^{\beta + \gamma - 1} E_{\beta, \beta + \gamma} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right), \end{aligned}$$

where $E_{a,b}(z)$ denotes the *two-parameter Mittag-Leffler function* [Pod99]:

$$E_{a,b}(z) := \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(ak + b)}, \quad a > 0, b > 0,$$

and Γ is the standard *Gamma-function*.

We mention that under the special case where $\alpha \in (1, 2)$, $\beta \in (0, 1]$ and $\gamma = 1 - \beta$, that $Z = Z^* = Y$, which is easily seen from their Fourier transforms given above. Because of these equalities, we simply let Y denote the fundamental solution under this situation. Moreover, under this case there is a probabilistic representation for Y which is used by the authors in [FLN19; FN17; MN15; FLO17; MN16]. Indeed, let X_t denote a symmetric α -stable process with density $p(t, x)$. Let $D = \{D_r, r \geq 0\}$ denote a β -stable subordinator and E_t its first passage time. Then it is known that the density of the time changed processes X_{E_t} is given by $Y(t, x)$ and we have that

$$Y(t, x) = \int_0^\infty p(s, x) f_{E_t}(s) ds,$$

where

$$f_{E_t}(s) = t^{\beta - 1} x^{-1 - 1/\beta} g_\beta \left(tx^{-1/\beta} \right),$$

where $g_\beta(\cdot)$ is the density function of D_1 and is smooth on the real line with $g_\beta(x) = 0$ for $x \leq 0$.

The goal of this paper is to prove the existence and uniqueness of a global solution to (1.1). The work is highly motivated by the recent work by Millet et al [MS21]. It is a well studied phenomena that either a super-linear drift or diffusion term may cause blow-up of the solution. As for the stochastic heat equation, we direct the reader to [FN21; MS93; DKZ19; BG09]. On the other hand, the only other work that we are aware of that is dedicated to proving some non-existence results of (1.1) is [AMN20]. So, to the best of our knowledge, this is the first work on proving the global existence of a solution to (1.1) with a super-linear diffusion term (see Theorem 1.1 below). However, as we will see, the calculations given below are essentially identical to those performed in [MS21] which is the cause for our motivation.

Here is the main result of this chapter. The proof will be postponed until Section 3 below.

C:MainSupLin

Theorem 1.1. *Suppose that σ satisfies (1.2) and the initial data are Borel measurable functions. Then for any $R > 0$ and $(t, x) \in [0, T] \times \bar{B}_R(0)$, where $\bar{B}_R(0)$ denotes the closed ball centered around the origin, there exists a random field solution to (1.1) denoted as $(u(t, x) : (t, x) \in [0, T] \times \bar{B}_R(0))$. This solution is unique and satisfies*

$$\sup_{t \in [0, T], |x| \leq R} |u(t, x)| < \infty, \quad \text{almost surely.} \quad (1.4)$$

The proof of Theorem 1.1 uses a standard stopping time argument along with a series of truncations of the super-linear term, σ , in order to construct a solution of (1.1). Indeed, we define

$$\sigma_N(x) = \begin{cases} \sigma(x) & |x| \leq N \\ \text{continuous and constant} & |x| > N, \end{cases}$$

and then associate for each $N > 0$, the corresponding global solution $u_N(t, x)$, whose existence follows due to Theorem 2.1 below. We then define the stopping time τ_N as follows:

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T.$$

We prove that $\{t < \tau_N\} \uparrow \Omega$ and then show that the solution to (1.1) exists pathwise for $\omega \in \Omega$. We remind the reader that in the case of a globally Lipschitz diffusion term, one usually proves that the solution exists as a limit in $L^2(\Omega)$. Because of the stopping-time argument, existence in $L^2(\Omega)$ is no longer guaranteed. Moreover, because $L^2(\Omega)$ existence is no longer guaranteed, then the solution may no longer satisfy the standard Ito isometry.

2 Preliminary results for a globally Lipschitz σ

In this section, we assume that the coefficient σ is globally Lipschitz and therefore satisfies the following inequality:

$$|\sigma(x)| \leq c(\sigma) + L(\sigma)|x|, \quad (2.1)$$

E:SupLinGlo

where $c(\sigma) = \sigma(0)$ and $L(\sigma)$ is the Lipschitz constant of σ and we assume that $L(\sigma) > 0$. The equation (1.1) under the assumption that σ satisfies (2.1) has been studied in [CHN19] where they establish in Theorem 4.1 *ibid* the existence and uniqueness of global random field solutions. In order to do so, one needs to show *Dalang's condition*:

$$\int_0^t ds \int_{\mathbb{R}^d} dy |Y(s, y)|^2 < \infty, \quad \text{for all } t > 0, \quad (2.2)$$

C:Dalang

which is equivalent to (e.g. see Lemma 5.3 *ibid*)

$$d < 2\alpha + \frac{\alpha}{\beta} \min\{2\gamma - 1, 0\} =: \Theta, \quad (2.3)$$

E:DalangEqu

which is further equivalent to

$$\rho(d) > 0 \quad \text{and} \quad d < 2\alpha, \quad (2.4) \quad \text{E: DalangEqu}$$

where

$$\rho(x) := 2\beta + 2\gamma - 1 - \beta x/\alpha. \quad (2.5) \quad \text{D: rho}$$

We now state the existence and uniqueness result proven in [CHN19].

1and2normYZ

Theorem 2.1. [CHN19, Theorem 4.1] Under (2.2), the SPDE (1.1) has a unique (in the sense of versions) random field solution $\{u(t, x) : (t, x) \in (0, \infty) \times \mathbb{R}^d\}$ if the initial data are such that

$$\widehat{C}_T := \sup_{t \in [0, T], x \in \mathbb{R}^d} |J_0(t, x)| < \infty. \quad (2.6) \quad \text{E: HomogBdd}$$

We now state a lemma proven in [CHN19] that calculates the $L^1(\mathbb{R}^d)$ and $L^2(\mathbb{R}^d)$ norms of the fundamental solutions. The lemma will be needed below in some of our calculations.

1and2normYZ

Lemma 2.2. [CHN19, Theorem 4.1 and Lemma 5.5] Assume that $d < 2\alpha$, $\beta \in (0, 2)$, and $\gamma \geq 0$. Then

$$\int_{\mathbb{R}^d} Y(t, x) dx = t^{\beta+\gamma-1} \quad \text{and} \quad \int_{\mathbb{R}^d} Z(t, x) dx = t^{\lceil \beta \rceil - 1}$$

and

$$\int_{\mathbb{R}^d} Y^2(t, x) dx = C_{\#} t^{2(\beta+\gamma-1)-d\beta/\alpha} = C_{\#} t^{\rho-1},$$

for all $t > 0$, where $\rho = \rho(d)$ and

$$C_{\#} := \frac{2}{\Gamma(d/2)(4\pi)^{d/2}} \int_0^\infty u^{d-1} E_{\beta, \beta+\gamma}^2(-u^\alpha) du.$$

Now further assume that $\beta \in (1, 2)$, then

$$\int_{\mathbb{R}^d} Z^*(t, x) dx = 1.$$

Remark 2.3. When $\alpha = \beta = 2$, $\gamma = 0$ and $d = 1$ then Lemma 2.2 gives us that $\int_{\mathbb{R}} Y^2(t, x) dx = t/2$ which coincides with the $L^2(\mathbb{R})$ norm of the wave kernel.

2.1 Some moment bounds

In this section, we will prove some moment bounds and a continuity result for the solution with a globally Lipschitz diffusion term. In the proof of the next proposition, we will use the following two facts:

$$\sup_{t \geq 0} t^k e^{-at} = k^k (ea)^{-k} \text{ for } k \in \mathbb{N} \text{ and } \sup_{t \geq 0} \int_0^t s e^{-as} ds = a^{-2} \text{ for } a > 0. \quad (2.7) \quad \text{E: SUPidenti}$$

P:N(u)Bdd

Proposition 2.4. [MS21, Proposition 3.2] Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$. Suppose that $d < 2\alpha$ and $\rho = \rho(d) > 0$, where $\rho(x)$ is defined in (2.5). Then there exists a universal constant $K := K(\alpha, \beta, \gamma, d) > 0$ such that for any

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (2.8) \quad \boxed{\text{E:Nbdd}}$$

where

$$\mathcal{T}_0 = \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \right]^p, \quad t \in [0, T]. \quad (2.9) \quad \boxed{\text{E:Pnormtbdd}}$$

Proof. First we fix an $a > 0$ and $p \in [2, \infty)$. Using Lemma 2.2 we see that,

$$\begin{aligned} |J(t, x)| &= \begin{cases} |(Z(t, \cdot) * u_0)(x)| & \beta \in (0, 1] \\ |(Z(t, \cdot) * v_0)(x) + (Z^*(t, \cdot) * u_0)(x)| & \beta \in (1, 2) \end{cases} \\ &\leq \begin{cases} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty & \beta \in (0, 1] \\ t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \end{aligned}$$

Now using (2.7), we see that

$$\begin{aligned} N_{a,p}(J) &\leq \begin{cases} \sup_{t \geq 0} \sup_{x \in \mathbb{R}} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty e^{-at} & \beta \in (0, 1] \\ \sup_{t \geq 0} \sup_{x \in \mathbb{R}} (t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty) e^{-at} & \beta \in (1, 2) \end{cases} \\ &= \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases} \quad (2.10) \quad \boxed{\text{E:Jbdd}} \end{aligned}$$

Now we find an upper bound for $N_{a,p}(I)$. By simultaneously applying the Burkholder-Davies-Gundy inequality and triangle inequality, we get that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 4p \left\| \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t - s, x - y) \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t - s, x - y) \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying Lemma 2.2 and the Lipschitz property of σ we get that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \int_0^t (t - s)^{\rho - 1} ds + 8pL^2(\sigma)N_{a,p}^2(u) \int_0^t ds \int_{\mathbb{R}^d} dy e^{2as} Y^2(t - s, x - y).$$

Now under Dalang's condition (2.4), the first integral above can be calculated as

$$\int_0^t (t - s)^{\rho - 1} ds = \frac{t^\rho}{\rho}.$$

By applying Lemma 2.2, we can calculate the second integral as

$$\int_0^t ds \int_{\mathbb{R}^d} dy e^{2as} Y^2(t - s, x - y) = C_\# \int_0^t ds e^{2as} (t - s)^{\rho - 1}.$$

Putting this together gives us that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \frac{t^\rho}{\rho} + 8pL^2(\sigma)N_{a,p}^2(u)C_\# \int_0^t ds e^{2as} (t - s)^{\rho - 1}.$$

Now by multiplying both sides by e^{-2at} , we see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_{\#}}{\rho} t^{\rho} e^{-2at} \\ &\quad + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#} \int_0^t ds e^{-2a(t-s)}(t-s)^{\rho-1}. \end{aligned}$$

We now do a change of variables and again recall the definition of the Gamma function to see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_{\#}}{\rho} t^{\rho} e^{-2at} + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#} \int_0^{\infty} du e^{-2au}(u)^{\rho-1} \\ &= \frac{8pc^2(\sigma)C_{\#}}{\rho} t^{\rho} e^{-2at} + 8pL^2(\sigma)N_{a,p}^2(u) \frac{C_{\#}\Gamma(\rho)}{(2a)^{\rho}} \end{aligned}$$

where the last line is true if Dalang's condition [CHN19] holds. If we now apply (2.7) and also recall that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, then we see that

$$N_{a,p}(I) \leq c(\sigma) \sqrt{\frac{8pC_{\#}}{\rho}} \left(\frac{\rho}{2ea}\right)^{\rho/2} + L(\sigma)N_{a,p}(u) \sqrt{\frac{8pC_{\#}\Gamma(\rho)}{(2a)^{\rho}}}. \quad (2.11) \quad \boxed{\text{E:SoleMil3.}}$$

To simplify things, we now let

$$K := \max \left\{ \sqrt{\frac{8C_{\#}}{\rho}} \left(\frac{\rho}{2e}\right)^{\rho/2}, \sqrt{\frac{8C_{\#}\Gamma(\rho)}{2^{\rho}}} \right\}$$

then,

$$N_{a,p}(I) \leq \frac{K\sqrt{p}}{a^{\rho/2}} (c(\sigma) + L(\sigma)N_{a,p}(u)). \quad (2.12) \quad \boxed{\text{E:Ibdd}}$$

Now by combining (2.10) and (2.12), we see that

$$N_{a,p}(u) \leq \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_{\infty} & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_{\infty} + \|u_0\|_{\infty} & \beta \in (1, 2). \end{cases} \quad (2.13)$$

$$+ K \frac{\sqrt{p}c(\sigma)}{a^{\rho/2}} + K \frac{\sqrt{p}L(\sigma)}{a^{\rho/2}} N_{a,p}(u). \quad (2.14) \quad \boxed{\text{E:Nbddgen}}$$

It is important to note that if we now restrict ourselves to the case of $\alpha = \beta = 2$, $d = 1$ and $\gamma = 0$, then (2.11) and (2.14) recover [MS21, (3.9) and (3.10)] respectively. Now choose $a > 0$ such that

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

This implies that the following interval is nonempty:

$$\left[2, \frac{a^{\rho}}{4L^2(\sigma)K^2} \right].$$

In addition, for any p in this interval, we have that

$$\frac{\sqrt{p}L(\sigma)K}{a^{\rho/2}} \leq \frac{1}{2},$$

and we also have

$$\frac{\sqrt{p}}{a^{\rho/2}} \leq \frac{1}{2KL(\sigma)}.$$

This along with (2.14) implies that

$$N_{a,p}(u) \leq 2 \begin{cases} ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|u_0\|_\infty & \beta \in (0, 1] \\ ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} + \frac{c(\sigma)}{L(\sigma)}$$

and this is precisely (2.8). Note that (2.9) follows immediately from (2.8). $\square$

We will need to recall the following result proved in [CHN19] in order to prove Proposition 2.7 below.

P:YInc

Proposition 2.5. [CH21, Proposition 2.1 and 2.2] *Suppose that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, and (2.3) holds. Consider $\rho = \rho(d)$ which is defined above in (2.5). Then $Y(t, x) = Y_{\alpha, \beta, \gamma, d}(t, x)$ satisfies the following:*

P:YIncCase1

1. *Suppose that $0 < \theta < (\Theta - d) \wedge 2$ where Θ is given in (2.3), $0 < t, r \leq T$ for some $T > 0$, $\beta \leq 1$ and $\gamma \leq [\beta] - \beta$. Then there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^\rho + |x-z|^\theta \right).$$

P:YIncCase2

2. *If we only assume that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, $0 < t, r \leq T$ for some $T > 0$ and (2.3) holds then for $0 < \theta < (\Theta - d) \wedge 2$ there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^q + |x-z|^\theta \right),$$

where

$$0 < q < \rho.$$

Remark 2.6. In Proposition 2.5 above, we may often ignore the dependence of C on $\alpha, \beta, \gamma, \nu, d$ and just write $C(\theta, T)$.

P:NormInc

Proposition 2.7. *Let $a > 0$ and suppose that $0 < \theta < (\Theta - d) \wedge 2$ and $\rho = \rho(d) > 0$. Then for any $x, z \in \mathbb{R}$, $p \geq 2$ and for any $T > 0$ with $t \in [0, T]$, the stochastic integral, $I(t, x)$, satisfies the following:*

$$\frac{\|I(t, x) - I(r, z)\|_p}{(|t-r|^q + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)], \quad (2.15)$$

E:NormInc_N

where

$$\mathcal{M}_1 = \sqrt{p}c(\sigma) \quad \text{and} \quad \mathcal{M}_2 = \sqrt{p}L(\sigma),$$

and

1. $0 < q \leq \rho$ under Case 1 of Proposition 2.5,
2. $0 < q < \rho$ under Case 2 of Proposition 2.5.

Moreover, for any

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right]$$

we have that

$$\frac{\|I(t, x) - I(r, z)\|_p}{(|t-r|^q + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) \left[\mathcal{M}_1 + \mathcal{M}_2 e^{aT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right) \right]. \quad (2.16)$$

E:NormInc_c

Proof. Note that because of the initial conditions, we see that $\|u(t, x) - u(r, z)\|_p = \|I(t, x) - I(r, z)\|_p$. We now apply Burkholder-Davies-Gundy inequality along with the triangle inequality, as was done in the proof of Proposition 2.4, to see that

$$\begin{aligned}\|I(t, x) - I(r, z)\|_p^2 &\leq 4p \left\| \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \|\sigma^2(u(s, y))\|_{p/2}.\end{aligned}$$

Now by applying the Lipchitz condition on σ gives us that

$$\begin{aligned}\|I(t, x) - I(r, z)\|_p^2 &\leq 8p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \left(c^2(\sigma) + L^2(\sigma) \|u(s, y)\|_p^2 \right) \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \exp(2as) N_{a,p}^2(u) \right] \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right].\end{aligned}$$

We now apply Proposition 2.5 to see that

$$\begin{aligned}\|I(t, x) - I(r, z)\|_p^2 &\leq 8p \left[c^2(\sigma) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right],\end{aligned}$$

where $C(\theta, T)$ is as in Proposition 2.5. Now by taking the square root of both sides and applying the identity $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ yields

$$\|I(t, x) - I(r, z)\|_p \leq 2\sqrt{2C(\theta, T)p} \left[c(\sigma) + L(\sigma) \exp(aT) N_{a,p}(u) \right] \left(|t-r|^q + |x-z|^\theta \right)^{1/2}.$$

This proves (2.15). It is clear that (2.16) follows directly from (2.15) and Proposition 2.4. $\square$

P:Holder

Proposition 2.8. *The stochastic integral, I , has a version, still denoted by I that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$ where $0 < \theta < (\Theta - d) \wedge 2$ and*

1. $0 < q \leq \rho$ under Case 1 of Proposition 2.5,
2. $0 < q < \rho$ under Case 2 of Proposition 2.5.

Moreover, if we consider

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right],$$

where K is the constant from Proposition 2.4, then

$$\mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] \leq 2^{p-1} \widehat{C}_T^p + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right],$$

where $\widehat{C}_T$ is defined in (2.6).

Proof. Proposition 2.7 implies that

$$\|I(t, x) - I(r, z)\|_p \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)] \left(|t - r|^q + |x - z|^\theta \right)^{1/2}. \quad (2.17)$$

By Kolmogorov's continuity theorem, there exists a version of I , which we denote by I , that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

Next, note that by triangle inequality,

$$\begin{aligned} |u(t, x)|^p &\leq 2^{p-1} (\|J_0(\cdot, \circ)\|_\infty^p + |u(t, x) - J_0(t, x)|^p) \\ &\leq 2^{p-1} \left(\widehat{C}_T^p + \sup_{t \in [0, T], |x| \leq R} |I(t, x)|^p \right) \\ &= 2^{p-1} \left(\widehat{C}_T^p + \sup_{t \in [0, T], |x| \leq R} |I(t, x)|^p \right), \end{aligned}$$

where we recall that $\widehat{C}_T$ defined in (2.6). Due to the continuity of I on the compact set $[0, T] \times \bar{B}_R(0)$, there exists a point $(t_0, x_0) \in [0, T] \times \bar{B}_R(0)$ such that

$$\sup_{t \in [0, T], |x| \leq R} |I(t, x)|^p = |I(t_0, x_0)|^p.$$

Thus, for any $(t, x) \in [0, T] \times \bar{B}_R(0)$,

$$|u(t, x)|^p \leq 2^{p-1} \left(\widehat{C}_T^p + |I(t_0, x_0)|^p \right)$$

which in return implies that

$$\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \leq 2^{p-1} \left(\widehat{C}_T^p + |I(t_0, x_0)|^p \right).$$

Now by taking the expectation of both sides and applying Proposition 2.7 gives us

$$\begin{aligned} \mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] &\leq 2^{p-1} \widehat{C}_T^p + 2^{p-1} C(p, \theta, T)^p \left((2T)^{q/2} + (2R)^{\theta/2} \right)^p [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)]^p \\ &= 2^{p-1} \widehat{C}_T^p + C(p, \theta, T, R) [\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} N_{a,p}(u)^p]. \end{aligned}$$

Now if we consider

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right],$$

then by Proposition 2.4, we have that

$$\mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] \leq 2^{p-1} \widehat{C}_T^p + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right].$$

□

Remark 2.9. The Hölder continuity in time proven above in Proposition 2.8 recovers the result proven by Foondun and Nane in [FN17, Theorem 1.9] if we restrict ourself to the case where $\beta \in (0, 1)$ and $\gamma = 1 - \beta$.

3 Proof of Theorem 1.1

Proof. We first solve a *truncated* version of (1.1). By this, we mean (1.1) with σ replaced by σ_N where in the case of $d = 1$,

$$\sigma_N(x) = \sigma(x)\mathbf{1}_{\{|x| \leq N\}} + \sigma(N)\mathbf{1}_{\{x > N\}} + \sigma(-N)\mathbf{1}_{\{x \leq -N\}},$$

and in higher dimensions σ_N is defined similarly. Note that since σ satisfies (1.2), then σ_N is Lipschitz continuous with Lipschitz constant $L(\sigma_N) := \sigma_2 \ln(2N)^\delta$. In other words,

$$|\sigma_N(x)| \leq \sigma_1 + \sigma_2 \ln(2N)^\delta |x|.$$

Hence we can apply Theorem 2.1 and see that there exists an unique solution of the truncated version of (1.1), which we denote by $u_N := \{u_N(t, x) : (t, x) \in [0, T] \times \mathbb{R}\}$. In addition, Proposition 2.8 implies that the solution has a version, still denoted by u_N , which is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

We will now apply Proposition 2.8 to find an upper bound on the p -norm of u_N . For this, consider

$$a > (8L^2(\sigma_N)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma_N)K^2}\right].$$

Then Proposition 2.8 implies that

$$\mathbb{E} \left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p \right) \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p(N) \exp(apT) \left(2 + \frac{\sigma_1}{L(\sigma_N)} \right) \right] \quad (3.1)$$

where we recall that

$$\mathcal{M}_1 = \sqrt{p}\sigma_1 \quad \text{and} \quad \mathcal{M}_2(N) = \sqrt{pL(\sigma_N)} = \sqrt{p\sigma_2 \ln(2N)^\delta}. \quad (3.2)$$

We now use the above to prove the existence and uniqueness of (1.1) with superlinear σ satisfying (1.2). For any $T > 0$ and $N \geq 2$ with $N \in \mathbb{N}$, we define the following stopping time:

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T. \quad (3.3)$$

The uniqueness of the solution and the local property of the stochastic integral imply that on $\{t < \tau_N\}$, $u_N(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. Hence, $(\tau_N)_{N \geq 2}$ is increasing and bounded above by T .

We now momentarily assume that $\sup_N \tau_N = T$ which would imply that $\{t < \tau_N\} \uparrow \Omega$. On each $\{t < \tau_N\}$, define $(u(t, x) : (t, x) \in [0, T] \times \mathbb{R})$ by $u(t, x) = u_N(t, x)$ and hence $u(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. This implies that on $\{t < \tau_N\}$

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma_N(u(s, y)) W(ds, dy).$$

However, on $\{t < \tau_N\}$, we have that $\sigma_N(u_N(t, x)) = \sigma(u(t, x))$ and so u satisfies

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad \{t < \tau_N\}.$$

Lastly, since $\{t < \tau_N\} \uparrow \Omega$, we conclude that

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad (t, x) \in [0, T] \times \mathbb{R}. \quad (3.4)$$

It is mentioned that the stochastic integrals are no longer defined in $L^2(\Omega)$. The extension of the integral is found in Dalang and Sanz-Sole's textbook that is not yet published.

We now focus on proving that $\sup_N \tau_N = T$, or equivalently that $\mathbb{P}(\tau_N < T) \rightarrow 0$ as $N \rightarrow \infty$. To do this, note that by Chebychev's inequality

$$\begin{aligned} \mathbb{P}(\tau_N < T) &\leq \mathbb{P}\left(\sup_{t \in [0, T], |x| \leq R} |u_N(t, x)| \geq N\right) \\ &\leq N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], |x| \leq R} |u_N(t, x)|^p\right). \end{aligned}$$

An application of (3.1) gives

$$\begin{aligned} N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], |x| \leq R} |u_N(t, x)|^p\right) &\leq N^{-p} \left(2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p\right]\right) \\ &= \frac{2^{p-1} + C(p, \theta, T, R) \mathcal{M}_1^p}{N^p} + \frac{C(p, T, R) \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p}. \end{aligned}$$

The first term clearly converges to 0 as $N \rightarrow \infty$. As for the second term, we expand using (3.2) to see that

$$\begin{aligned} \frac{\mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p} &= \frac{\left(\sqrt{p\sigma_2 \ln(2N)^\delta}\right)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{\sqrt{p\sigma_2 \ln(2N)^\delta}}\right)^p}{N^p} \\ &\sim \frac{\ln(2N)^{p\delta/2}}{N^p} \rightarrow 0, \end{aligned}$$

where above we use the symbol $\sim$ to mean asymptotically equivalent as $N \rightarrow \infty$. This completes the proof of the Theorem. $\square$

4 Some comments and current progress with a spatially colored noise

Consider equation 1.1 driven by a noise that is white in time and colored space. In other words, W is a 0-mean Gaussian process indexed on the space of Schwartz functions with the following covariance structure:

$$C(\phi, \psi) := \int_0^\infty dt \int_{\mathbb{R}^d} \Gamma(dx) (\phi(t, \cdot) * \tilde{\psi}(t, \cdot))(x), \quad \tilde{\psi}(x) = \psi(-x). \quad (4.1) \quad \boxed{\text{E:Colored}}$$

As per usual, the tempered measure Γ is non-negative and nonnegative definite and satisfies the following three conditions:

NoiseAssum1

Assumption 4.1.

1. the spectral measure μ is such that $\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} < \infty$,
2. Γ is absolutely continuous with respect to the Lebesgue measure and therefore has a density, $\Gamma(dx) = f(x)dx$. Furthermore, when this is the case, we may often write $\mu(d\zeta) = \varphi(\zeta)d\zeta$ where we have the relation $\varphi = \mathcal{F}f$,
3. the spectral density, φ , satisfies the following scaling property: $\varphi(c\xi) = c^{-(d-\lambda)}\varphi(\xi)$ for some $\lambda > 0$.

Remark 4.2. To the best of our knowledge, the only correlation function and spectral density that satisfies the conditions of Assumption 4.1 is the Riesz kernel with parameter $0 < \Lambda < d$,

$$f(x) = \frac{1}{|x|^\Lambda},$$

and its fractional generalization,

$$f(x) = \prod_{i=1}^k \frac{1}{|x_{(i)}|^{\Lambda_i}},$$

for $0 < \Lambda_i < d_i$ where $\sum_{i=1}^k d_i = d$ and $\sum_{i=1}^k \Lambda_i =: \Lambda \in (0, d)$ and $x_{(i)} = (x_1^i, \dots, x_{d_i}^i)$ where $x_{(i)}$ essentially partitions the coordinate $x = (x_1, \dots, x_d)$.

RieszKernel

Assumption 4.3. Because of the above remark, we will assume that the correlation function is given by the Riesz kernel. In other words, for $\lambda \in (0, d)$:

$$f(x) := \frac{1}{|x|^\lambda} \quad \text{and} \quad \varphi(x) := \mathcal{F}f(\xi) = \frac{K_f}{|\xi|^{d-\lambda}},$$

where

$$K_f = \pi^{-d/2} 2^{-\lambda} \frac{\Gamma((d-\lambda)/2)}{\Gamma(\lambda/2)}.$$

Taking into consideration the first point from Assumption 4.1, we must take

$$0 < d - \Lambda < d \quad \text{and} \quad d - \Lambda + 2\alpha > d,$$

and since $\Lambda \in (0, d)$, this implies that

$$0 < \Lambda < d \quad \text{and} \quad \Lambda < 2\alpha,$$

or in other words,

$$0 < \Lambda < \min\{d, 2\alpha\}.$$

Under Assumption 4.1, we may rewrite (4.1) as

$$C(\phi, \psi) = \int_0^\infty dt \int_{\mathbb{R}^{2d}} dx dy \phi(t, x) f(x - y) \psi(t, y)$$

and by applying Placherel's identity, this further implies that

$$\int_0^\infty dt \int_{\mathbb{R}^{2d}} dx dy \phi(x) f(x - y) \psi(y) = (2\pi)^{-d} \int_0^\infty dt \int_{\mathbb{R}^{2d}} \mu(d\xi) \mathcal{F}(\phi(t, \cdot))(\xi) \overline{\mathcal{F}(\psi(t, \cdot))(\xi)}.$$

We now prove a lemma which is a generalization of equation (2.2) which accounts for the colored noise.

L:YfYbdd

Lemma 4.4. Suppose Assumption 4.1 holds. Then there exists a constant $C(\beta, \gamma) =: C > 0$ such that for any $x, y \in \mathbb{R}^d$ we have that

$$\int_{\mathbb{R}^{2d}} Y(t, x - z) Y(t, y - w) f(w - z) dz dw \leq C \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_{\mathbb{R}^d} \frac{\mu(d\xi)}{(1 + |\xi|^\alpha)^2} < \infty \quad (4.2)$$

E:YYfBdd

Proof. Through a change of variables, we may write the left side of (4.2) as the following:

$$\int_{\mathbb{R}^{2d}} Y(t, x - z) Y(t, y - w) f(w - z) dz dw = \int_{\mathbb{R}^d} Y(t, u - (x - y)) (Y(t, \cdot) * f)(u) du,$$

where $*$ denotes the spatial convolution. Now an application of Parseval's identity gives us that

$$\int_{\mathbb{R}^d} Y(t, u - (x - y)) (Y(t, \cdot) * f)(u) du = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}Y(t, \cdot - (x - y))(\zeta) \mathcal{F}(Y(t, \cdot) * f)(\zeta) d\zeta.$$

Applying the translation property of the Fourier transform and [CHN19, Theorem 4.1] gives us that

$$\begin{aligned} \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}Y(t, \cdot - (x - y))(\zeta) \mathcal{F}(Y(t, \cdot) * f)(\zeta) d\zeta \\ = \frac{t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \exp(-i(x - y)\zeta) E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\zeta|^\alpha}{2} \right)^2 \mu(d\zeta). \end{aligned}$$

Hence by taking the absolute value on both sides and recalling the positivity of f , Y and the Mittag-Leffler function we get that

$$\int_{\mathbb{R}^{2d}} Y(t, x - z) Y(t, y - w) f(w - z) dz dw \leq \frac{t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\zeta|^\alpha}{2} \right)^2 \mu(d\zeta).$$

Now, by applying [Pod99, Theorem 1.6], we see that there exists a $C > 0$ depending only on β and γ such that

$$\frac{t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\zeta|^\alpha}{2} \right)^2 \mu(d\zeta) \leq \frac{C t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + \frac{\nu}{2} t^\beta |\zeta|^\alpha)^2}.$$

Another change of variables implies that

$$\frac{C t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + \frac{\nu}{2} t^\beta |\zeta|^\alpha)^2} \leq \frac{C}{(2\pi)^d} \left(\frac{\nu}{2} \right)^{-\lambda/\alpha} t^{2(\beta+\gamma-1) - \beta\lambda/\alpha} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + |\zeta|^\alpha)^2}.$$

The finiteness of the above integral follows from Assumption 4.1. $\square$

We will now prove existence and uniqueness of the solution of (1.1) driven by a noise that is white in time and colored in space and when we take σ to be globally Lipschitz,

$$|\sigma(x) - \sigma(y)| \leq L(\sigma)|x - y|,$$

or if we take $c(\sigma) = \sigma(0)$,

$$|\sigma(x)| \leq c(\sigma) + L(\sigma)|x|.$$

ExisAndUniq

Theorem 4.5. *Suppose that $\lambda\beta/\alpha - 2(\beta + \gamma - 1) < 1$ and in addition, suppose that $\lambda < 2\alpha$. Then there exists a unique $L^2(\Omega)$ -solution to (1.1) driven by $\dot{F}$ such that*

$$u(t, x) = J_0(t, x) + I(t, x), \quad a.s.$$

Proof. The proof follows the strategy laid out in Walsh [Wal86]. We start by defining

$$u_0(t, x) := J_0(t, x) \quad \text{and} \quad u_n(t, x) := J_0(t, x) + \int_0^t \int_{\mathbb{R}^d} Y(t - s, x - y) \sigma(u_n(s, y)) F(ds, dy).$$

In addition, we define

$$D_n(t, x) := \mathbb{E} |u_{n+1}(t, x) - u_n(t, x)|^2 \quad \text{and} \quad H_n(t) := \sup_{x \in \mathbb{R}^d} D_n(t, x).$$

Lastly, we define

$$\Sigma(t, x, n) := |\sigma(u_n(t, x)) - \sigma(u_{n-1}(t, x))|.$$

With this notation we see that

$$D_n(t, x) = \int_0^t ds \int_{\mathbb{R}^d} dz dy Y(t-s, x-y) Y(t-s, x-z) f(y-z) \mathbb{E}(\Sigma(s, y, n) \Sigma(s, z, n)).$$

Now note that by the Lipschitz property of σ , we have that

$$\begin{aligned} \mathbb{E}(\Sigma(s, y, n) \Sigma(s, z, n)) &\leq L_\sigma^2 \mathbb{E}(|u_n(s, y) - u_{n-1}(s, y)| |u_n(s, z) - u_{n-1}(s, z)|) \\ &\leq L_\sigma^2 \|u_n(s, y) - u_{n-1}(s, y)\| \|u_n(s, z) - u_{n-1}(s, z)\| \\ &\leq L_\sigma^2 \sqrt{D_{n-1}(s, y) D_{n-1}(s, z)} \\ &\leq L_\sigma^2 H_{n-1}(s). \end{aligned}$$

Thus,

$$\begin{aligned} D_n(t, x) &\leq L_\sigma^2 \int_0^t H_{n-1}(s) \int_{\mathbb{R}^d} dz dy Y(t-s, x-y) Y(t-s, x-z) f(y-z) \\ &\leq CL_\sigma^2 \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} \int_0^t ds \frac{H_{n-1}(s)}{(t-s)^{\lambda\beta/\alpha - 2(\beta+\gamma-1)}}, \end{aligned}$$

where the last inequality follows from Lemma 4.4. In return, this implies that

$$H_n(t) \leq CL_\sigma^2 \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} \int_0^t ds \frac{H_{n-1}(s)}{(t-s)^{\lambda\beta/\alpha - 2(\beta+\gamma-1)}}.$$

Since we assume that $\lambda\beta/\alpha - 2(\beta+\gamma-1) < 1$, then the result follows from the discussion before the proof of [Wal86, Lemma 3.3]. $\square$

Remark 4.6. We must strengthen Assumptions 4.1 and 4.3 because of the need to require $\lambda\beta/\alpha - 2(\beta+\gamma-1) < 1$ in Theorem 4.5. More precisely, we require that

$$\lambda < \frac{2\alpha}{\beta}(\beta+\gamma-1) + \frac{\alpha}{\beta} = \frac{\alpha}{\beta} [2(\beta+\gamma)-1].$$

Therefore, Assumption 4.3 to require that

$$\Lambda < \min \left\{ d, 2\alpha, \frac{\alpha}{\beta} [2(\beta+\gamma)-1] \right\}, \quad (4.3) \quad \boxed{\text{A:RieszPara}}$$

which is the same condition that was assumed in [CE22, Theorem 1.6] where they studied (1.1) but with a time-independent noise.

We will now focus on proving some moment bounds that parallel those proved above in the space-time white noise case.

4.1 Some moment bounds for a globally Lipschitz σ

Proposition 4.7. [MS21, Proposition 4.4] Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$. Suppose that $d < 2\alpha$ and for this proof consider $\rho = \rho(\lambda)$ which is defined in (??). Then there exists a universal constant $K := K(\alpha, \beta, \gamma, d, \lambda) > 0$ such that for any

$$a^\rho > (8L^2(\sigma)K^2).$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (4.4) \quad \boxed{\text{E:ColorNbdd}}$$

where

$$\mathcal{T}_0 = \begin{cases} ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|u_0\|_\infty & \beta \in (0, 1] \\ ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \right]^p, \quad t \in [0, T]. \quad (4.5) \quad \boxed{\text{E:ColorPMom}}$$

Proof. We note that $\mathcal{N}_{a,p}(J)$ has the same upper bound as in Proposition 2.4 above. In other words,

$$\mathcal{N}_{a,p}(J) \leq \begin{cases} \|u_0\|_\infty & \beta \in (0, 1] \\ (ea)^{-1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

We now work to find an upper bound for $\mathcal{N}_{a,p}(I)$. The Burkholder-Davies-Gundy inequality and Hölder's inequality imply that

$$\|I(t, x)\|_p^2 \leq 4p \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) \|\sigma(u(s, y_1))\|_p \|\sigma(u(s, y_2))\|_p.$$

We now apply the Lipschitz property to see that for $i \in \{1, 2\}$,

$$\|\sigma(u(s, y_i))\|_p \leq c(\sigma) + L(\sigma) \|u(s, y_i)\|_p.$$

Therefore, we may say that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 4p \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) \\ &\quad \times \prod_{i=1}^2 c(\sigma) + L(\sigma) \|u(s, y_i)\|_p \\ &\leq 4p \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) [c(\sigma) + L(\sigma) \exp(as) N_{a,p}(u)]^2 \\ &\leq 8pc(\sigma)^2 \int_0^t ds \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2) \\ &\quad + 8pN_{a,p}(u)^2 L(\sigma)^2 \int_0^t ds \exp(2as) \iint_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, x-y_1) f(x-y) Y(t-s, x-y_2). \end{aligned}$$

Now by applying Lemma 4.4, we may bound the $\iint_{\mathbb{R}^{2d}} dy_1 dy_2$ integrals by the quantity

$$K_1(t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha},$$

where

$$K_1 := CK_f \left(\frac{\nu}{2} \right)^{-\lambda/\alpha},$$

and C is as in Lemma 4.4. In return, this gives us that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 8pK_1 c(\sigma)^2 \int_0^t (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} ds \\ &\quad + 8pK_1 N_{a,p}(u)^2 L(\sigma)^2 \int_0^t \exp(2as) (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} ds \end{aligned}$$

$$\begin{aligned}
&= 8pK_1c(\sigma)^2 \frac{t^\rho}{\rho} + 8pK_1N_{a,p}(u)^2 L(\sigma)^2 \frac{\exp(2at) [\Gamma(\rho) - \Gamma(\rho, 2at)]}{(2a)^\rho} \\
&=: \Delta_1(t) + \Delta_2(t),
\end{aligned}$$

where $\Gamma(\cdot)$ denotes the Gamma function and $\Gamma(\cdot, \circ)$ is the incomplete-upper Gamma function. Note that

$$\mathcal{N}_{a,p}(I)^2 \leq \sup_{t \geq 0} \exp(-2at) (\Delta_1(t) + \Delta_2(t)).$$

We start with bounding $\sup_{t \geq 0} \exp(-2at) \Delta_1(t)$. Recall the identities in (2.7). With this we see that

$$\sup_{t \geq 0} \exp(-2at) t^\rho = (\rho)^\rho (2ea)^{-\rho}.$$

Hence,

$$\sup_{t \geq 0} \exp(-2at) \Delta_1(t) \leq \frac{8pK_1c(\sigma)^2}{\rho} \left(\frac{\rho}{2ea} \right)^\rho.$$

Next we bound $\sup_{t \geq 0} \exp(-2at) \Delta_2(t)$. Note that for any $t \geq 0$,

$$\begin{aligned}
\exp(-2at) \Delta_2(t) &= 8pK_1N_{a,p}(u)^2 L(\sigma)^2 \frac{[\Gamma(\rho) - \Gamma(\rho, 2at)]}{(2a)^\rho} \\
&\leq 8pK_1N_{a,p}(u)^2 L(\sigma)^2 \frac{\Gamma(\rho)}{(2a)^\rho}.
\end{aligned}$$

Putting this all together gives us that

$$\mathcal{N}_{a,p}(I)^2 \leq \frac{8pK_1c(\sigma)^2}{\rho} \left(\frac{\rho}{2ea} \right)^\rho + \frac{8pK_1\Gamma(\rho)L(\sigma)^2}{(2a)^\rho} N_{a,p}(u)^2,$$

and thus,

$$\mathcal{N}_{a,p}(I) \leq 2c(\sigma) \sqrt{\frac{2pK_1}{\rho}} \left(\frac{\rho}{2ea} \right)^{\rho/2} + 2L(\sigma) \frac{\sqrt{2pK_1\Gamma(\rho)}}{(2a)^{\rho/2}} N_{a,p}(u).$$

Similar to the proof of Proposition 2.4 above, we define

$$K = K(\alpha, \beta, \gamma, d, \lambda) := \max \left\{ 2\sqrt{\frac{2K_1}{\rho}} \left(\frac{\rho}{2e} \right)^{\rho/2}, 2\frac{\sqrt{2K_1\Gamma(\rho)}}{(2)^{\rho/2}} \right\}.$$

We now see that

$$\mathcal{N}_{a,p}(I) \leq Kc(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} + KL(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} \mathcal{N}_{a,p}(u) \quad (4.6)$$

This and the bound for $\mathcal{N}_{a,p}(J)$ found above implies that

$$\begin{aligned}
\mathcal{N}_{a,p}(u) &\leq \mathcal{N}_{a,p}(J) + \mathcal{N}_{a,p}(I) \\
&\leq \begin{cases} \|u_0\|_\infty & \beta \in (0, 1] \\ (ea)^{-1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \\
&\quad + Kc(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} + KL(\sigma) \frac{\sqrt{p}}{a^{\rho/2}} \mathcal{N}_{a,p}(u).
\end{aligned} \quad (4.7)$$

Now consider

$$a^\rho > 8K^2 L(\sigma)^2.$$

This implies that the interval

$$\left[2, \frac{a^\rho}{4K^2 L(\sigma)^2} \right]$$

is nonempty. Hence, for any p in this interval we have that

$$\frac{KL(\sigma)\sqrt{p}}{a^{\rho/2}} \leq \frac{1}{2}.$$

This along with (4.7) implies that

$$\begin{aligned} \mathcal{N}_{a,p}(u) \leq 2 & \begin{cases} \|u_0\|_\infty & \beta \in (0, 1] \\ (ea)^{-1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \\ & + \frac{c(\sigma)}{L(\sigma)}. \end{aligned}$$

Its easy to see that (4.5) follows directly from (4.4). $\square$

We now state a lemma which will be used later on.

Parseval

Lemma 4.8. *Let ϕ and ψ be bounded Borel functions. Then for $s, t > 0$ and $z \in \mathbb{R}^d$ we have,*

$$\iint_{\mathbb{R}^{2d}} dx dy \phi(x) Y(t, x) \psi(y) Y(s, y) f(x - y + z) = (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}(\phi Y(t, \cdot))(\xi) \overline{\mathcal{F}(\psi Y(s, \cdot))(\xi)} e^{-iz \cdot \xi} \mu(d\xi) \quad (4.8)$$

Proof. For notation, define $Y_\phi(t, x) = \phi(x) Y(t, x)$ and $Y_\psi(t, x) = \psi(x) Y(t, x)$. The boundedness and measurability of ϕ and ψ guarantees the existence of $\mathcal{F}Y_\phi$ and $\mathcal{F}Y_\psi$. Thus we have,

$$\iint_{\mathbb{R}^{2d}} dx dy Y_\phi(t, x) Y_\psi(s, y) f(x - y + z) = \int_{\mathbb{R}^d} dx Y_\phi(t, x) (Y_\psi(s, \cdot) * f(\cdot + z))(x).$$

Parseval's identity implies that

$$\begin{aligned} \int_{\mathbb{R}^d} dx Y_\phi(t, x) (Y_\psi(s, \cdot) * f(\cdot + z))(x) &= (2\pi)^{-d} \int_{\mathbb{R}^d} d\zeta \mathcal{F}(Y_\phi(t, \cdot))(\zeta) \overline{\mathcal{F}((Y_\psi(s, \cdot))(\zeta) \mathcal{F}(f(\cdot + z)))} \\ &= (2\pi)^{-d} \int_{\mathbb{R}^d} \mu(d\zeta) \mathcal{F}(Y_\phi(t, \cdot))(\zeta) \overline{\mathcal{F}((Y_\psi(s, \cdot))(\zeta))} e^{-iz \cdot \zeta}, \end{aligned}$$

where $\mu(d\xi) = \varphi(\xi) d\xi$. This completes the proof. $\square$

Proposition 4.9. [MS21, Proposition 4.8] *Statement...*

Proof. The following proof follows closely the techniques presented in [HHN14, Theorem 3.1]. By the Burkholder Davies Gundy inequality, we have that

$$\begin{aligned} \|I(t, x) - I(t, z)\|_p^2 &\leq 4p \left\| \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 f(y_1 - y_2) \prod_{i=1}^2 (Y(t - s, x - y_i) - Y(t - s, z - y_i) \sigma(u(s, y_i))) \right\|_{p/2} \\ &=: 4p \|Q\|_{p/2} \end{aligned}$$

We now reorganize the integrand of Q above by following [HHN14, Theorem 3.1]. By expanding the integrand of Q , we get that

$$\begin{aligned} Q &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, x - y_1) Y(t - s, x - y_2) f(y_1 - y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \\ &\quad - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, x - y_1) Y(t - s, z - y_2) f(y_1 - y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \\ &\quad - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t - s, z - y_1) Y(t - s, x - y_2) f(y_1 - y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \end{aligned}$$

$$\begin{aligned}
& + \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, z-y_1) Y(t-s, z-y_2) f(y_1-y_2) \sigma(u(s, y_1)) \sigma(u(s, y_2)) \\
& =: I_1 + I_2 + I_3 + I_4.
\end{aligned}$$

Now let $\xi = x - z$ and consider the following notation:

$$\begin{aligned}
\Sigma_x(s, y) &= \sigma(u(s, x-y)), \quad \Sigma_{x,z}(s, y) = \sigma(u(s, x-y)) - \sigma(u(s, z-y)), \\
h_1(s, y_1, y_2) &= f(y_1-y_2) \Sigma_{x,z}(s, y_1) \Sigma_{x,z}(s, y_2) \\
h_2(s, y_1, y_2) &= [f(y_1-y_2+\xi) - f(y_1-y_2)] \Sigma_x(s, y_2) \Sigma_{x,z}(s, y_1) \\
h_3(s, y_1, y_2) &= h_2(s, y_2, y_1) \\
h_4(s, y_1, y_2) &= [2f(y_1-y_2) - f(y_1-y_2+\xi) - f(y_1-y_2-\xi)] \Sigma_x(s, y) \Sigma_x(s, z).
\end{aligned}$$

With this notation along with a change of variables, we rewrite I_1 through I_4 as the following:

$$\begin{aligned}
I_1 &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1-y_2) \Sigma_x(s, y_1) \Sigma_x(s, y_2) \\
I_2 &= - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1-y_2-\xi) \Sigma_x(s, y_1) \Sigma_z(s, y_2) \\
&= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1-y_2-\xi) \Sigma_x(s, y_1) \Sigma_{x,z}(s, y_2) \\
&\quad - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1-y_2-\xi) \Sigma_x(s, y_1) \Sigma_x(s, y_2) \\
I_3 &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1-y_2+\xi) \Sigma_x(s, y_2) \Sigma_{x,z}(s, y_1) \\
&\quad - \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1-y_2+\xi) \Sigma_x(s, y_1) \Sigma_x(s, y_2) \\
I_4 &= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1-y_2) \Sigma_z(s, y_1) \Sigma_z(s, y_2) \\
&= \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1-y_2) (\Sigma_{x,z}(s, y_1) - \Sigma_x(s, y_1)) (\Sigma_{x,z}(s, y_2) - \Sigma_x(s, y_2)),
\end{aligned}$$

where above I_3 is derived in the same way I_2 was derived. Now if we expand I_4 and regroup all the terms in $I_1 + \dots + I_4$, we get that

$$Q = \sum_{i=1}^4 Q_i \quad (4.9)$$

where

$$Q_i = \int_0^t \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) h_i(s, y_1, y_2). \quad (4.10)$$

Thus

$$\|I(t, x) - I(t, z)\|_p^2 \leq 4p \sum_{i=1}^4 \|Q_i\|_{p/2}. \quad (4.11)$$

We now estimate each $\|Q_i\|_{p/2}$ separately.

Upper bound for $\|Q_1\|_{p/2}$. Using the Lipschitz property of σ , Minkowski's inequality and the triangle inequality gives is

$$\begin{aligned}
\|Q_1\|_{p/2} &\leq L(\sigma)^2 \int_0^t ds \int_{\mathbb{R}^{2d}} dy_1 dy_2 Y(t-s, y_1) Y(t-s, y_2) f(y_1-y_2) \left[\sup_{|z_1-z_2|=\xi} \|u(s, z_1) - u(s, z_2)\|_p^2 \right] \\
&\leq L(\sigma)^2 C K_f \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} \int_0^t ds (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \left[\sup_{|z_1-z_2|=\xi} \|u(s, z_1) - u(s, z_2)\|_p^2 \right]
\end{aligned}$$

$$\leq L(\sigma)^2 CK_f \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} (t)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_0^t ds \left[\sup_{|z_1-z_2|=\xi} \|u(s, z_1) - u(s, z_2)\|_p^2 \right],$$

where the second inequality above follows from Lemma 4.4. last line requires $2(\beta + \gamma - 1) - \beta\lambda/\alpha > 0$. If we leave the $(t - s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha}$ term in the integral then we may be able to assume the weaker condition that $2(\beta + \gamma - 1) - \beta\lambda/\alpha > -1$, thus not excluding the SHE.

To bound the remaining $Q'_i s$, we will need to consider the following truncations which will then allow us to apply Lemma 4.8. For $k \geq 1$ we define $\Sigma_x^k(s, y) = \Sigma_x(s, y) 1_{\{\Sigma_x(s, y) \leq k\}}$ and $\Sigma_{x,z}^k(s, y) = \Sigma_{x,z}(s, y) 1_{\{\Sigma_{x,z}(s, y) \leq k\}}$ and then we define Q_i^k the same as Q_i but with h_i replaced with h_i^k where h_i^k is defined by replacing $\Sigma_x(s, y)$ and $\Sigma_{x,z}(s, y)$ with $\Sigma_x^k(s, y)$ and $\Sigma_{x,z}^k(s, y)$ respectively in h_i .

Upper Bound for $\|Q_2^k\|_{p/2}$. We apply Lemma 4.8 with $\phi(y_2) = \Sigma_x^k(s, y_2)$ and $\psi(y_1) = \Sigma_{x,z}^k(s, y_1)$. Thus we have that up to a constant of $(2\pi)^{-d}$,

$$Q_2^k = \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) \overline{\mathcal{F}(\Sigma_x^k(s, \cdot) Y(t-s, \cdot))(\zeta)} \mathcal{F}(\Sigma_{x,z}^k Y(t-s, \cdot))(\zeta) [e^{-i\xi \cdot \zeta} - 1],$$

where we recall that $\xi = x - z$. We now apply the the following to bounds: $|e^{-i\xi \cdot \zeta} - 1| \leq C|\xi|^r |\zeta|^r$ for any $r \in [0, 1]$ and some $C > 0$ and $\sqrt{ab} \leq \frac{1}{2}(a+b)$ for and $a, b > 0$. Indeed, for some $r \in [0, 1]$,

$$\begin{aligned} |Q_2^k| &\leq C \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) \left| \mathcal{F}(\Sigma_x^k(s, \cdot) Y(t-s, \cdot))(\zeta) \right| \left| \mathcal{F}(\Sigma_{x,z}^k Y(t-s, \cdot))(\zeta) \right| |\xi|^r |\zeta|^r \\ &\leq \int_0^t ds |\xi|^r \left(\int_{\mathbb{R}^d} \mu(d\zeta) \left| \mathcal{F}(\Sigma_x^k(s, \cdot) Y(t-s, \cdot))(\zeta) \right|^2 |\zeta|^{2r} \right)^{1/2} \\ &\quad \times \left(\int_{\mathbb{R}^d} \mu(d\zeta) \left| \mathcal{F}(\Sigma_{x,z}^k Y(t-s, \cdot))(\zeta) \right|^2 \right)^{1/2} \\ &\leq \frac{1}{2} \int_0^t ds |\xi|^{2r} \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} \left| \mathcal{F}(\Sigma_x^k(s, \cdot) Y(t-s, \cdot))(\zeta) \right|^2 \\ &\quad + \int_{\mathbb{R}^d} \mu(d\zeta) \left| \mathcal{F}(\Sigma_{x,z}^k(s, \cdot) Y(t-s, \cdot))(\zeta) \right|^2 \\ &=: Q_{2,1}^k + Q_{2,2}^k. \end{aligned}$$

As for $Q_{2,1}^k$,

$$\begin{aligned} \|Q_{2,1}^k\|_{p/2} &\leq C|\xi|^{2r} \left\| \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \Sigma_{x,z}^k(s, y) \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \right\|_{p/2} \\ &\leq C|\xi|^{2r} \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \left\| \Sigma_{x,z}^k(s, y) \right\|_p^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\ &\leq C|\xi|^{2r} \left[c(\sigma) + L(\sigma) \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \end{aligned}$$

which follows from the global Lipschitz assumption on σ . Now as for $Q_{2,2}^k$,

$$\begin{aligned} \|Q_{2,2}^k\|_{p/2} &\leq \int_0^t ds \sup_{y \in \mathbb{R}^d} \left\| \Sigma_{x,z}^k(s, y) \right\|_p^2 \int_{\mathbb{R}^d} \mu(d\zeta) |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\ &\leq C \int_0^t ds \sup_{y \in \mathbb{R}^d} \left\| \Sigma_{x,z}^k(s, y) \right\|_p^2 (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} \end{aligned}$$

¹There is a comment made in [MS21, Proposition 4.8] and [HHN14, Theorem 3.2] that minkowski inequality is being used for the first inequality but it seems incorrect how they apply minkowski's and also unnecessary.

$$\begin{aligned}
&\leq C \left(\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} \right) L^2(\sigma) \int_0^t ds (t-s)^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \sup_{|z_1-z_2|=|\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2 \\
&\leq C \left(\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} \right) L^2(\sigma) t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_0^t ds \sup_{|z_1-z_2|=|\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2,
\end{aligned}$$

where in the second to last line we applied Lemma 4.4 and in the last line we assume that $2(\beta + \gamma - 1) - \beta\lambda/\alpha \geq 0$. Summarizing the above gives us that

$$\begin{aligned}
\|Q_2^k\|_{p/2} &\leq C|\xi|^{2r} \left[c(\sigma) + L(\sigma) \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\
&\quad + C \left(\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} \right) L^2(\sigma) t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_0^t ds \sup_{|z_1-z_2|=|\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2.
\end{aligned}$$

Also by swapping x and z and noticing that $|e^{-i\xi\zeta} - 1| = |e^{i\xi\zeta} - 1|$ then the bound for $\|Q_3^k\|_{p/2}$ with be the same as for $\|Q_2^k\|_{p/2}$.

Lastly, for Q_4^k we have that after applying Lemma 4.8 with $\phi = \psi = \Sigma_x^k(s, \cdot)$,

$$\begin{aligned}
|Q_4^k| &\leq C \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) \left| 2 - e^{-i\xi\zeta} - e^{i\xi\zeta} \right| \left| \mathcal{F} \left(\Sigma_x^k(s, \cdot) Y(t-s, \cdot) \right) (\zeta) \right|^2 \\
&\leq C|\xi|^{2r} \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} \left| \mathcal{F} \left(\Sigma_x^k(s, \cdot) Y(t-s, \cdot) \right) (\zeta) \right|^2 \\
&= C|\xi|^{2r} \int_0^t ds \int_{\mathbb{R}^d} dy g(y) \left[\left(\Sigma_x^k(s, \cdot) Y(t-s, \cdot) \right) * \left(\widetilde{\Sigma_x^k(s, \cdot) Y(t-s, \cdot)} \right) \right] (y),
\end{aligned}$$

where $g = \mathcal{F}^{-1}(\mu(\cdot) |\cdot|^{2r})$. Thus we see that

$$\begin{aligned}
\|Q_4^k\|_{p/2} &\leq C|\xi|^{2r} \int_0^t ds \sup_{y \in \mathbb{R}^d} \left\| \Sigma_x^k(s, y) \right\|_p^2 \int_{\mathbb{R}^d} dy g(y) \left[(Y(t-s, \cdot)) * (\widetilde{Y(t-s, \cdot)}) \right] (y) \\
&= C|\xi|^{2r} \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \left\| \Sigma_x^k(s, y) \right\|_p^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\
&\leq C|\xi|^{2r} \left[c(\sigma) + L(\sigma) \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2.
\end{aligned}$$

Thus taking (4.11) into consideration, we have that

$$\begin{aligned}
\|I(t, x) - I(t, z)\|_p^2 &\leq C|\xi|^{2r} \left[c(\sigma) + L(\sigma) \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\
&\quad + C \left(\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} \right) L^2(\sigma) t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_0^t ds \sup_{|z_1-z_2|=|\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2.
\end{aligned}$$

Lastly we note that by the same techniques in Lemma 4.4, we see that

$$\begin{aligned}
&\int_0^t ds \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} |\mathcal{F}(Y(t-s, \cdot))(\zeta)|^2 \\
&\leq \int_0^t ds (t-s)^{2(\beta+\gamma-1)} \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} E_{\beta, \beta+\gamma} \left(-2^{-1} \nu(t-s)^\beta |\zeta|^\alpha \right)^2 \\
&\leq C \int_0^t ds (t-s)^{2(\beta+\gamma-1)} \int_{\mathbb{R}^d} \mu(d\zeta) |\zeta|^{2r} \left(1 + 2^{-1} \nu(t-s)^\beta |\zeta|^\alpha \right)^{-2}
\end{aligned}$$

$$= C \int_0^t ds (t-s)^{2(\beta+\gamma-1)-\frac{\beta\lambda}{\alpha}-\frac{2r\beta}{\alpha}} \int_{\mathbb{R}^d} \mu(d\zeta) \frac{|\zeta|^{2r}}{(1+|\zeta|^\alpha)^2},$$

therefore,

$$\begin{aligned} & \|I(t, x) - I(t, z)\|_p^2 \\ & \leq C|\xi|^{2r} \left[\int_{\mathbb{R}^d} \mu(d\zeta) \frac{|\zeta|^{2r}}{(1+|\zeta|^\alpha)^2} \right] \left[\int_0^t ds (t-s)^{2(\beta+\gamma-1)-\frac{\beta\lambda}{\alpha}-\frac{2r\beta}{\alpha}} \right] \left[c(\sigma) + L(\sigma) \sup_{(s,y) \in [0,T] \times \mathbb{R}^d} \|u(s, y)\|_p \right]^2 \\ & \quad + C \left[\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\zeta|^\alpha)^2} \right] L^2(\sigma) t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_0^t ds \sup_{|z_1-z_2|=|\xi|} \|u(s, z_1) - u(s, z_2)\|_p^2, \end{aligned}$$

which requires us to assume that

$$2(\beta + \gamma - 1) - \frac{\beta\lambda}{\alpha} > 0 \quad \text{and} \quad 2(\beta + \gamma - 1) - \frac{\beta\lambda}{\alpha} - \frac{2r\beta}{\alpha} > -1$$

□

References

- blow-up:20** [AMN20] Sunday A. Asogwa, Jebessa B. Mijena, and Erkan Nane. “Blow-up results for space-time fractional stochastic partial differential equations”. In: *Potential Anal.* 53.2 (2020), pp. 357–386 (cit. on p. 3).
- global:09** [BG09] J Bonder and P. Groisman. “Space time white noise and non-Lipschitz reaction term”. In: *Physica D* (2009), pp. 271–215 (cit. on p. 3).
- interpolating** [CE22] Le Chen and Nicholas Eisenberg. “Interpolating the Stochastic Heat and Wave Equations with Time-independent Noise: Solvability and Exact Asymptotics”. In: *Stoch. Partial Differ. Equ. Anal. Comput. (pending revision) preprint at arXiv:2108.11473* (2022) (cit. on p. 14).
- holder:21** [CH21] Le Chen and Guannan Hu. “Hölder regularity of the nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$ ”. In: *Preprint arXiv:2105.00891* (2021) (cit. on p. 7).
- nonlinear:19** [CHN19] Le Chen, Yaozhong Hu, and David Nualart. “Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$ ”. In: *Stochastic Process. Appl.* 129.12 (2019), pp. 5073–5112 (cit. on pp. 2–4, 6, 7, 13).
- global:19** [DKZ19] Robert C. Dalang, Davar Khoshnevisan, and Tusheng Zhang. “Global solutions to stochastic reaction-diffusion equations with super-linear drift and multiplicative noise”. In: *Ann. Probab.* 47.1 (2019), pp. 519–559 (cit. on p. 3).
- osgood:21** [FN21] M Foondun and E Nualart. “The Osgood condition for stochastic partial differential equations”. In: (2021), pp. 295–311 (cit. on p. 3).
- some:19** [FLN19] Mohammud Foondun, Wei Liu, and Erkan Nane. “Some non-existence results for a class of stochastic partial differential equations”. In: *J. Differential Equations* 266.5 (2019), pp. 2575–2596 (cit. on p. 2).
- moment:17** [FLO17] Mohammud Foondun, Wei Liu, and McSylvester Omaba. “Moment bounds for a class of fractional stochastic heat equations”. In: *Ann. Probab.* 45.4 (2017), pp. 2131–2153 (cit. on p. 2).
- asymptotic** [FN17] Mohammud Foondun and Erkan Nane. “Asymptotic properties of some space-time fractional stochastic equations”. In: *Math. Z.* 287.1-2 (2017), pp. 493–519 (cit. on pp. 2, 9).

ing.ea:14:on	[HHN14]	Yaozhong Hu, Jingyu Huang, and David Nualart. “On Hölder continuity of the solution of stochastic wave equations in dimension three”. In: <i>Stoch. Partial Differ. Equ. Anal. Comput.</i> 2.3 (2014), pp. 353–407 (cit. on pp. 17, 19).
:space-time	[MN15]	Jebessa B. Mijena and Erkan Nane. “Space-time fractional stochastic partial differential equations”. In: <i>Stochastic Process. Appl.</i> 125.9 (2015), pp. 3301–3326 (cit. on p. 2).
termittence	[MN16]	Jebessa B. Mijena and Erkan Nane. “Intermittence and space-time fractional stochastic partial differential equations”. In: <i>Potential Anal.</i> 44.2 (2016), pp. 295–312 (cit. on p. 2).
e:21:global	[MS21]	Annie Millet and Marta Sanz-Solé. “Global solutions to stochastic wave equations with superlinear coefficients”. In: <i>Stochastic Process. Appl.</i> 139 (2021), pp. 175–211 (cit. on pp. 3, 4, 6, 14, 17, 19).
s:93:blowup	[MS93]	C Mueller and R Sowers. “Blowup for the heat equation with a noise term”. In: <i>Probability Theory and Related Fields</i> 97 (1993), pp. 287–320 (cit. on p. 3).
:fractional	[Pod99]	Igor Podlubny. <i>Fractional differential equations</i> . Vol. 198. An introduction to fractional derivatives, fractional differential equations, to methods of their solution and some of their applications. Academic Press, Inc., San Diego, CA, 1999, pp. xxiv+340. ISBN: 0-12-558840-2 (cit. on pp. 2, 13).
ntroduction	[Wal86]	John B. Walsh. “An introduction to stochastic partial differential equations”. In: <i>École d’été de probabilités de Saint-Flour, XIV—1984</i> . Vol. 1180. 1986, pp. 265–439 (cit. on pp. 2, 13, 14).